

BONE PATHOLOGY V OSTEOMYELITIS

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I. OSTEOMYELITIS

Definition:

Osteomyelitis is generally accepted as an inflammatory process of **bone and bone marrow** caused by **an infectious organism(s)** .

Bacteria reach the bone

- **hematogenous** spread from distant focus
- Direct spread from **adjacent soft tissue, compound fracture or prosthetic joint**
- **Ischemic , diabetic osteomyelitis**

Etiology

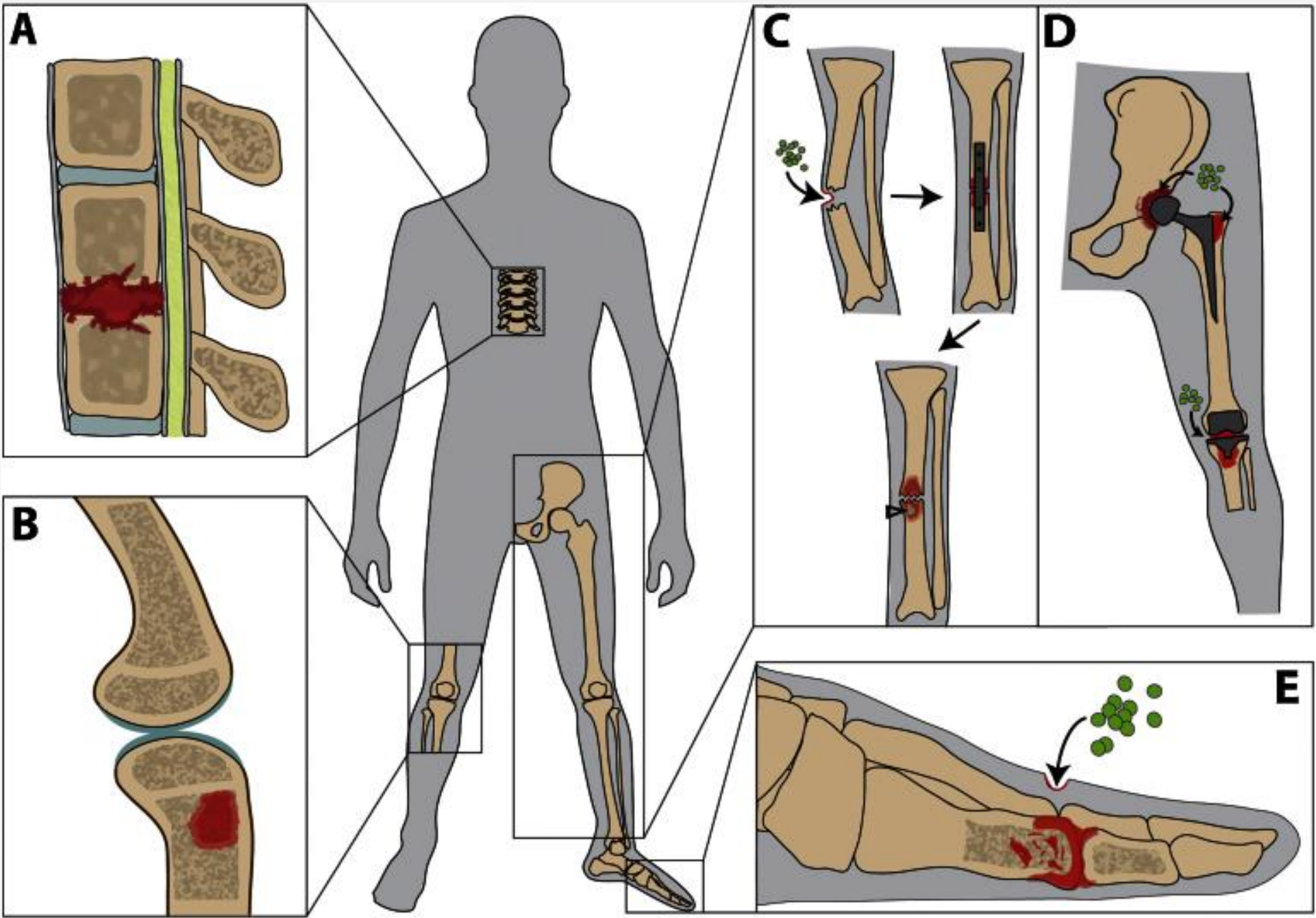
Hematogenous osteomyelitis

- occurs more frequently in children compared to adults,
- long bones are usually affected
- In adults, hematogenous osteomyelitis affects the vertebrae most commonly

Contiguous osteomyelitis

- in the setting of trauma and related surgery,
- related to decubitus ulcers and infected joint arthroplasties

any abnormal blood supply - worry about infection that can spread



Case I

A previously healthy **18-month-old boy** presented at the emergency department with left hip pain and a limp **following a minor trauma**.

His mother reported that he had presented fever for three days, **cough and rhinitis** about 15 days before the trauma, and had been treated with ibuprofen .

The child presented with a limited and **painful range of motion** of the left hip and could not bear weight on that side. Examination of the other **joints was unremarkable**, and no inflammatory signs were evidenced.

Blood tests were performed and showed white blood cell (WBC): 23,000 cells/ μ L; neutrophils: 86%, C-reactive protein (CRP): 7.2 mg/dL; erythrocyte sedimentation rate (ESR): 52 mm/h. Blood cultures were taken at admission,

Differential diagnosis

Osteomyelitis is known as the “**great pretender**” because of the difficulties encountered in diagnosis.

- malignant and benign tumors,
- cysts
- old trauma,
- bone infarcts secondary to sickle cell anemia, or other
- hemoglobinopathies.

Causative organisms

- 70-80% **Staphylococcus aureus**
 - Newborns: E. coli, Group B Strep
 - IV drug users: Pseudomonas
 - Sick cell patients Salmonella
- **Staphylococcus epidermidis** = medical devices
- polymicrobial = gram-negative bacteria (**Enterobacteriaceae**) resist a wide range of antibiotics.

***Early and specific
treatment is
important***

PATHOPHYSIOLOGY

Understanding of the pathophysiology of osteomyelitis provides insight into why **chronicity** and the presence of an **implant** require more vigorous treatment

Important factors in the pathogenesis of osteomyelitis

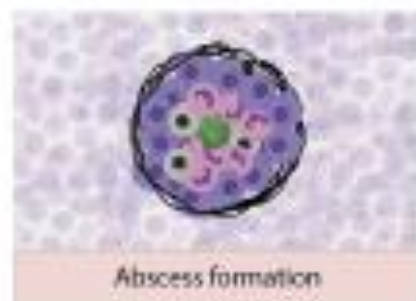
- the **virulence** of the **infecting organism**, (MRSA)
- underlying **disease and immune** status of the **host**,
- type, location, and vascularity of the **bone**

Bacteria factors :

1. **Staphylococcal abscess community (SAC)** surrounded by immune cells
2. **Biofilm** adhering to implanted devices: Bacteria in biofilms are less susceptible to antibiotics,
3. **S. aureus can survive within an osteocyte canaliculus**
these canalicular networks may be impenetrable by immune cells, and bacteria can possibly survive in this space for a long period of time,
4. **S. aureus** bacterium within a PMN
5. **S. aureus** has a wide range of toxins that target host cells.: exfoliative toxins, pore-forming toxins, and superantigens

A

Bacteria



Location, and vascularity of the bone

Hematogenous osteomyelitis occurs most often in **prepubertal children** and usually involves the metaphysis of long bones, particularly the **tibia and femur**.

In adults, the most common site is the **vertebral bodies**, followed by **long bones, pelvis, and clavicle**.

The primary blood supply of the vertebrae is the segmental arteries, which divide to perfuse segments of **two adjacent vertebrae**. Thus, vertebral osteomyelitis often occurs in **two contiguous vertebral bodies** and the intervertebral disc.

sharing of blood supply

MORPHOLOGY

Subperiosteal abscess

Sequestrum

Draining sinuses

Involucrum

Brodie's abscess

Morphology:

Acute: When bone tissue is infected, the bacteria induce an acute inflammatory reaction. The bacteria and inflammation affect the ***periosteum*** and *spread within the bone* causing bone necrosis.

In children, the periosteum is loosely attached to the cortex, allowing for the formation of sizable **subperiosteal abscesses** along the bone surface.

Rupture of a subperiosteal abscess may lead to a soft-tissue abscess and the eventual formation of a **draining sinus**.

Lifting of the periosteum further impairs the blood supply to the affected bone causing segmental bone necrosis known as a **sequestrum**.

Morphology

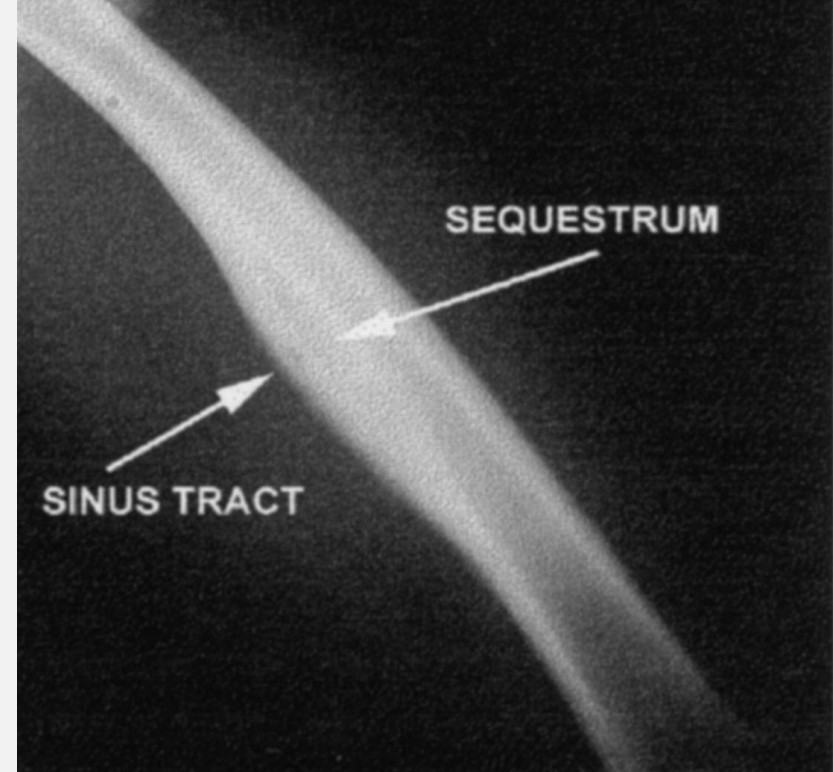
Chronic:

- Osteoclastic bone **resorption**,
- ingrowth of **fibrous tissue**,
- deposition of **reactive new** bone in the periphery.

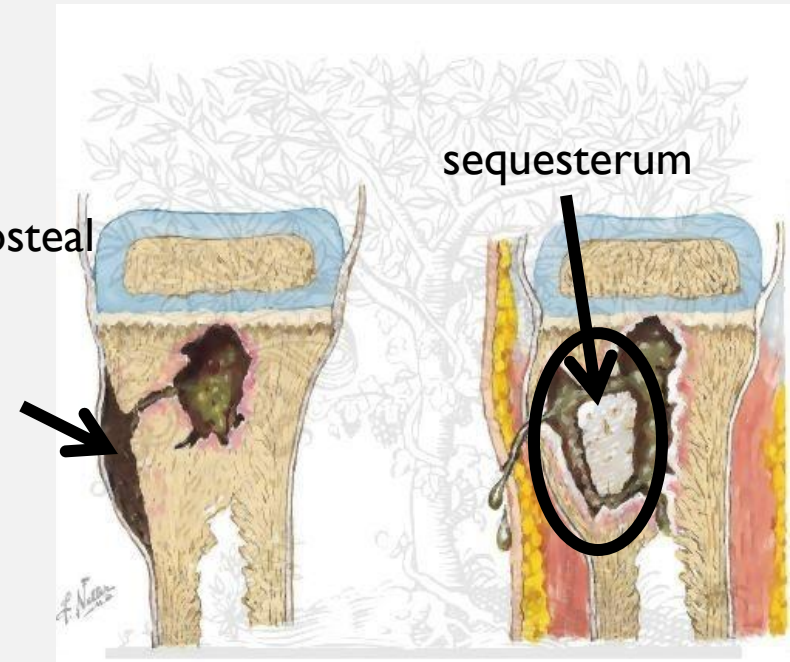
When the newly deposited bone forms a sleeve of living tissue around the segment of devitalized infected bone, it is known as an **involucrum**.

Acute: In 48 hrs after infection, inflammation extend through the Haversian system to reach the periosteum forming **sub periosteal abscess**.

The necrotic bone is known as **sequestrum**

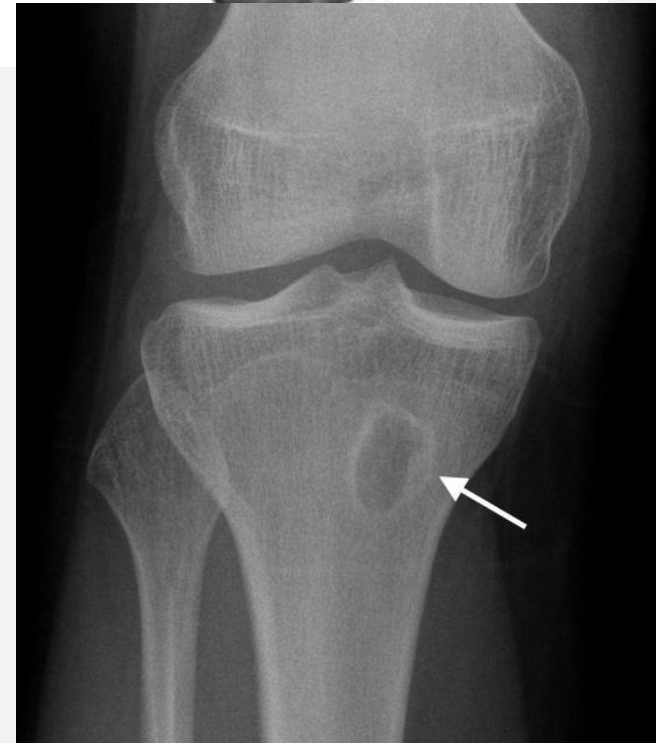
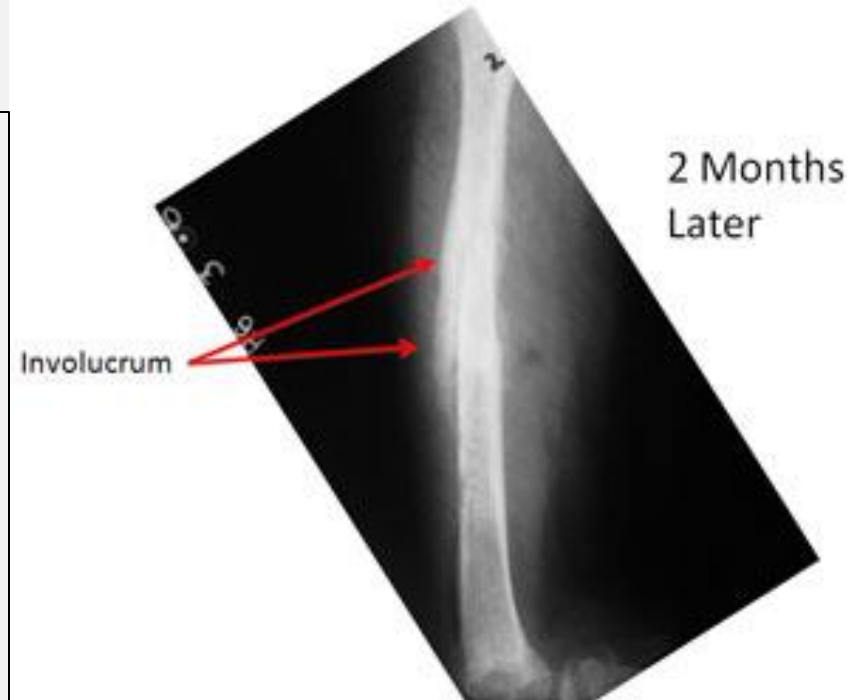


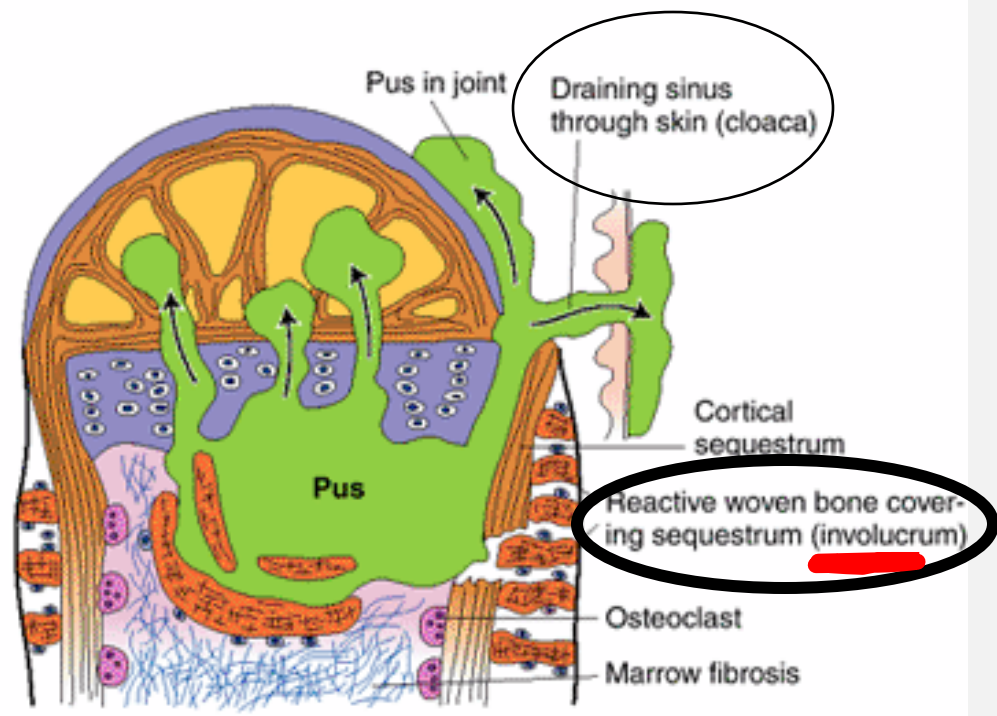
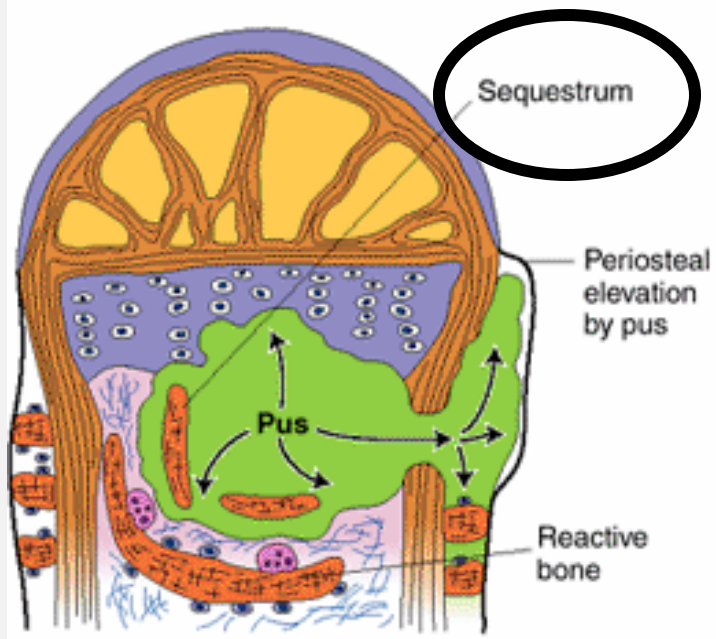
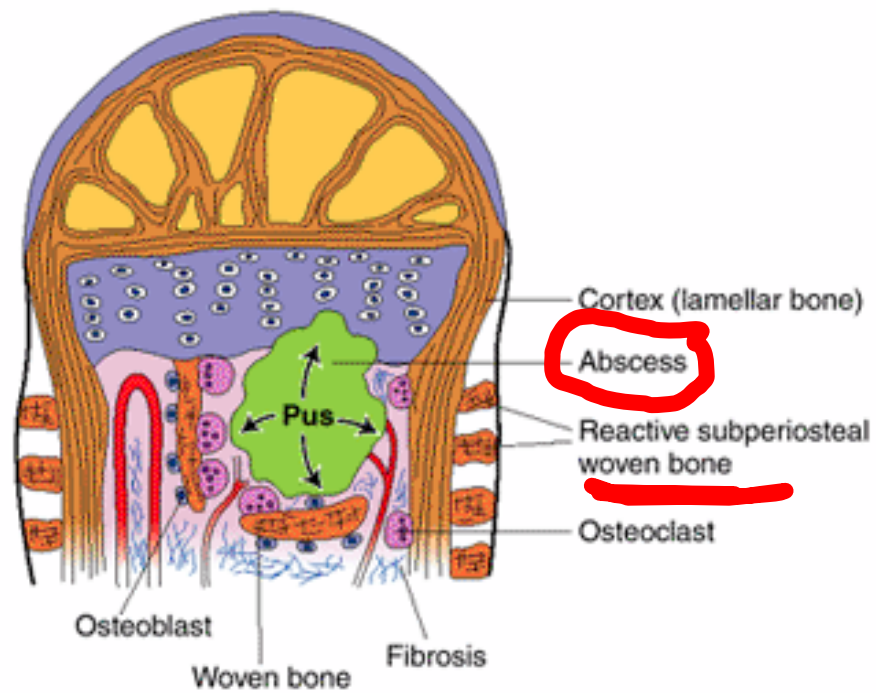
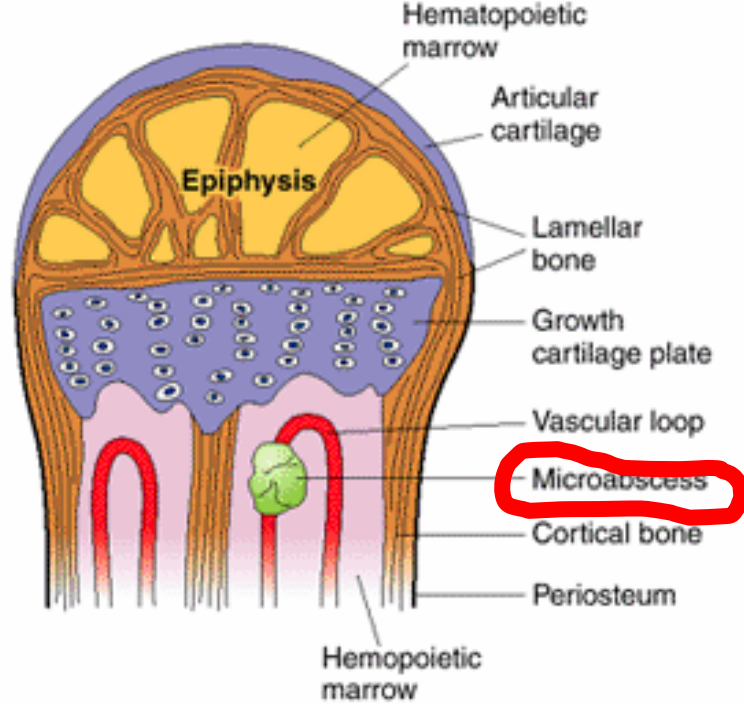
Subperiosteal abscess



Chronic: After several weeks, the reactive **newly** developed known **Involucrum** develops .

Brodie abscess low-grade pyogenic abscesses of the bone surrounded by reactive bone

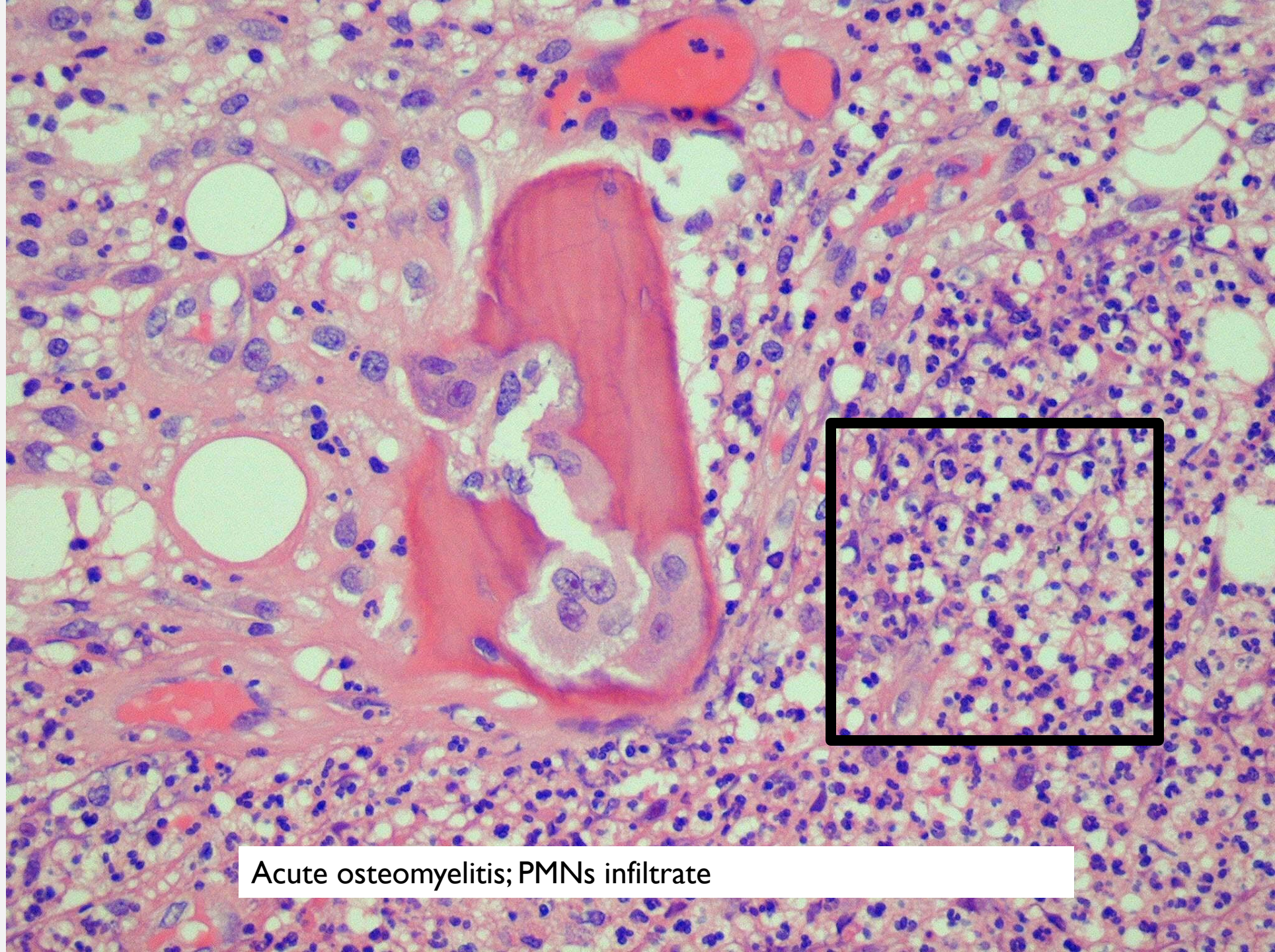




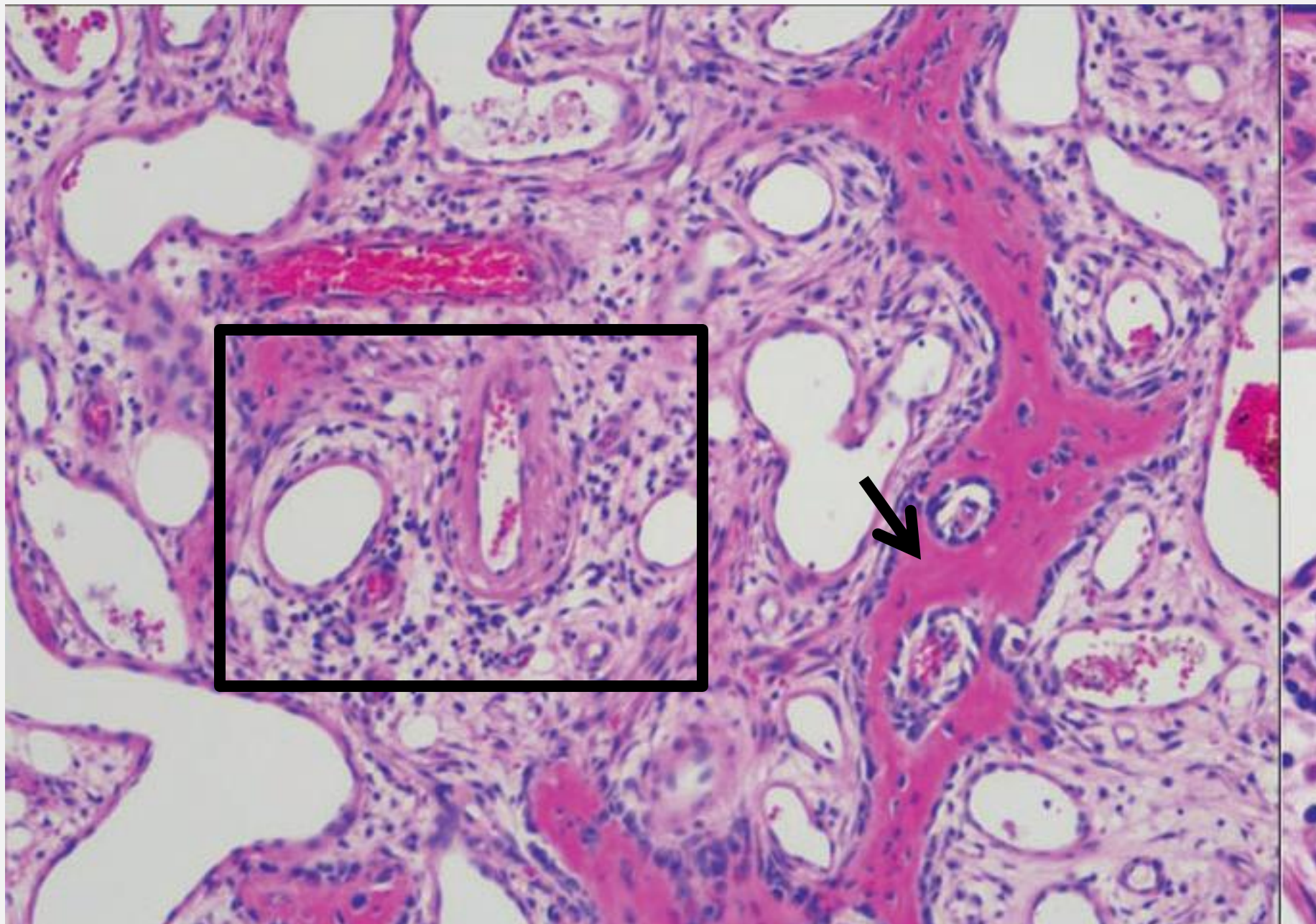
Lab Investigations

- Raised inflammatory markers such as
 - **C-reactive protein (CRP)**
 - **Erythrocyte sedimentation rate (ESR)**
- The gold standard for diagnosis is positive microbial cultures from bone biopsies and is the most sensitive diagnostic tool.
- Tissue samples should undergo both microbiological and histopathological analysis.

Radiographic imaging is an essential component of the evaluation of a patient with suspected osteomyelitis.



Acute osteomyelitis; PMNs infiltrate



Chronic osteomyelitis: new bone **formation** (**black arrow**), and replacement of normal bone marrow with **fibrovascular** inflammatory tissue (box square)

TUBERCULOSIS OSTEOMYELITIS

Spine (Pott disease)

- thoracic and lumbar vertebrae
- breaks through intervertebral
- soft tissue abscess
- Knees, hips

